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SAMPLE

Standard documentation which customers receive

1) Catalogue
- The catalogues for equipment which is used in switchgear
2) Configurations
- Source file for configuration of protection relays
3) Drawings
- Electrical drawings of switchgear (see below)
4) Protection and control
-Operation and technical manual for protection relay
5) Routine test reports

Structure of ABB documentations (According to 1VLD110063)

Basic submittal
- General specification
- Single Line Diagram - SLD
- General Arrangement - GA
- Foundation frame - FF (could be included in GA)
- Interconnection and Interlocking - ICL
- Information pages - INF

General drawings
- The drawings for each typical panel: Single line diagram, connections, low voltage compartment layout, door layout, terminals diagram....

Reference designation of functional units in switchgear
M - 1 kV - 6 kV
L - 6 kV - 10 kV
K - 10kV - 20 kV
J - 20 kV - 30 kV
number means designation of functional unit
Additional ABB Letter code A, B, C ...
ABB Standard 2nba000001

Internal number of project

Revision table
Rev.Code: A, B, C, D ....
Date:
Name:
Remark:
ABB standard 1VLD110063

Internal number for each set of drawings (typical panel)
ABB Standard PN0002

Name of customer, if is specified
Customer logo, if is specified

LV - low voltage compartment
CT - current transformers
LCR - LV - right side
LVD - LV door
.
.
.
ABB Standard 1VLD11037

Customer project and document number, if is specified

Name of drawing, for example: Current transformers

DCC code and Page
ABB Stand. 2NDA000001

SAMPLE

Drawn by	Checked by	Released by
name	name	name

Revision	Date	Name	Remark

Project number	1098888888
Document number	1VL84000001
Date	date

ABB s.r.o.
EPDS Brno

Customer project number	
Customer document number	
Customer	

Project name	12 kV Switchgear Sample
Page description	COMMON DRAWINGS ABB Plotframe

Installation =	K00	
Place of installation +		
DCC & &EFS	Page A00	Pages 5

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SAMPLE

ABB s.r.o., EPDS BRNO
Videnska 117
CZ-619 00 BRNO

12 kV SWITCHGEAR TYPE UNIGEAR ZS1

CUSTOMER: Customer name
CUSTOMER REF. No.:
ABB REF. No.: 1098888888

SWITCHGEAR BASIC DATA

STANDARD	IEC 62271-200
RATED VOLTAGE	12 kV
SERVICE VOLTAGE	12 kV
RATED FREQUENCY	50 Hz
POWER FREQUENCY WITHSTAND VOLTAGE / BIL	28 kV / 75 kV
RATED BUSBAR CURRENT	1250 A
RATED SHORT-TIME WITHSTAND CURRENT / MAKE CURRENT	31.5 kA / 80 kA
RATED DURATION OF SHORT CIRCUIT	1 s
RATED SHORT-TIME WITHSTAND CURRENT FOR EARTHING CIRCUITS	31.5 kA
RATED DURATION OF SHORT CIRCUIT FOR EARTHING CIRCUIT	1 s
INTERNAL ARC CLASSIFICATION (IAC) TYPE	AFLR
INTERNAL ARC CLASSIFICATION (IAC) CURRENT	31.5 kA
INTERNAL ARC CLASSIFICATION (IAC) TIME	1 s
DEGREE OF PROTECTION	IP4X
DEGREE OF PROTECTION - INTERNAL	IP20
MV PHASE DESIGNATION	L1, L2, L3
AMBIENT TEMPERATURE	-5 °C ... 40 °C
ALTITUDE	≤1000 m

ABB PROJECT REALIZATION TEAM

PROJECT MANAGER:	name
PROJECT ENGINEER:	name
DESIGN ENGINEER:	name

DURING THE INSTALLATION, COMMISSIONING PERIOD AND THE TIME OF OPERATION, CLIENT IS OBLIGED TO FOLLOW ALL INSTRUCTIONS AND RECOMMENDATION GIVEN BY MANUFACTURERS OF INDIVIDUAL INSTRUMENTS INSTALLED INSIDE SUPPLIED EQUIPMENT

SAMPLE

Drawn by	Checked by	Released by
name	name	name

Revision	Date	Name	Remark

Project number	1098888888
Document number	1VL84000000
Date	date

ABB s.r.o.
EPDS Brno

Customer project number
Customer document number
Customer

Project name: 12 kV Switchgear Sample
Page description: COMMON DRAWINGS
PROJECT TITLE PAGE

Installation = K00
Place of installation +

DCC &	Page	Pages
&EAA	A01	5

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SAMPLE

SHEET INDEX

PAGE	PAGE NAME	REVISION		
=K00/A01	PROJECT TITLE PAGE			
=K00/A02	SHEET INDEX			
=K00/C01	GEN. SPECIFICATION OF SWGR.			
=K00/C02	GEN. SPECIFICATION OF SWGR.			

SAMPLE

Drawn by	Checked by	Released by
name	name	name

Revision	Date	Name	Remark

Project number	1098888888
Document number	1VL84000000
Date	date

ABB s.r.o.
EPDS Brno

Customer project number
Customer document number
Customer

Project name: 12 kV Switchgear Sample

Page description: COMMON DRAWINGS
SHEET INDEX

Installation = K00

Place of installation +

DCC &	Page	Pages
	A02	5

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SAMPLE

GENERAL INFORMATION

SWITCHGEAR TYPE	UniGear Single busbar system
PACKING	Wooden pallet

SECONDARY WIRES PARAMETERS

STANDARD		ABB EPDS Brno Standard 1VLD110036
AC CIRCUIT	PHASE	BK (Black) / 1.5 mm²
	NEUTRAL	LB (Light blue) / 1.5 mm²
DC CIRCUIT	POSITIVE	BK (Black) / 1.5 mm²
	NEGATIVE	BK (Black) / 1.5 mm²
CT SECONDARY CIRCUIT	PHASE 1	BK (Black) / 2.5 mm²
	PHASE 2	BK (Black) / 2.5 mm²
	PHASE 3	BK (Black) / 2.5 mm²
	COMMON	BK (Black) / 2.5 mm²
VT SECONDARY CIRCUIT	PHASE 1	BK (Black) / 1.5 mm²
	PHASE 2	BK (Black) / 1.5 mm²
	PHASE 3	BK (Black) / 1.5 mm²
	NEUTRAL	LB (Light blue) / 1.5 mm²
EARTHING CIRCUIT		GNYE (Green-yellow) / 2.5 mm²
SUPPLY INTERCONNECTION		4 mm²
LOGIC INTERCONNECTION		2.5 mm²
SECONDARY WIRES TERMINATION		Non insulated crimping or welding
MARKING OF WIRE ENDS (MATERIAL)		Terminal (Slimfix) *)
SEC. WIRES TYPE AND NOMINAL VOLTAGE		PVC - Flame retardant, 0.45/0.75 kV

AUXILIARY VOLTAGES

CONTROL/SIGNALIZATION/PROTECTION CIRCUITS	2 DC 110 V, 6 kA
CIRCUIT BREAKER SPRING CHARGER MOTOR	2 DC 110 V, 6 kA
CIRCUIT BREAKER TRUCK MOTOR	-
EARTHING SWITCH MOTOR DRIVE	-
2nd TRIPPING COIL	2 DC 110 V, 6 kA
INTERNAL LIGHTING AND HEATING	1/N AC 230 V, 50 Hz, 6 kA
AUXILIARY VOLTAGES CONNECTION POSITION	Leftmost panel

OTHER ELECTRICAL PARAMETERS

TERMINALS TYPE / PRODUCER	Spring type / ABB
FAULT LIMITING DEVICE	1x Ith Limiter for every comp. (3 pcs)
ANTICONDENS. HEATER IN COMP. LV/CB/CABLE	Yes/Yes/Yes
INTERNAL LIGHTING	LV compartment light with IR sensor
REMOTE COMMUNICATION PROTOCOL	IEC 61850 Single
IED COMMUNICATION PROTOCOL 1st / 2nd	IEC 61850 Single / None/2.0
COMMUNICATION MEDIA	Galvanic (RJ45)

SAMPLE

*) BECAUSE OF TECHNICAL REASONS, MARKING FOR THE INTERCONNECTION WIRES (I.E. FOR THE BUNDLES OF WIRES BETWEEN PANELS) WILL BE OF THE TERMINAL TYPE ON WHITE RING FERRULES.

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SAMPLE

SWITCHGEAR MECHANICAL DESIGN

COATING COLOR	RAL 7035
Cu BARS DESIGN	
MAIN CIRCUIT Cu BARS COATING	No **)
BUSBAR PARTITIONING	Yes - according to internal standard
INSULATED T-OFFS AND BUSBARS	No ***)
LOW VOLTAGE COMPARTMENT	
LV COMPARTMENT DOORS DESIGN	Standard
MIMIC DIAGRAM	Not required
MIMIC DIAGRAM COLOR	N/A
LV ASSEMBLY PANEL MECH. DESIGN	Aluminium profiles
LOCKS	
LV COMPARTMENT DOORS CLOSING VERSION	T-Handle Double Bit 3.2 mm
BREAKER COMPARTMENT DOORS CLOSING VERS.	Handle Double Bit 5.2 mm
CABLE COMPARTMENT DOORS CLOSING VERSION	Handle Double Bit 5.2 mm
LABELS	
UNIT NAMEPLATES LANGUAGE	English
UNIT NAMEPLATES DESIGN	Engraved - fixed by plastic rivet
UNIT NAMEPLATES COLOR	Black letters/white background
DEVICES ON LV DOORS LABELS LANGUAGE	English
DEVICES ON LV DOORS LABELS DESIGN	Engraved – fixed by plastic rivet
DEVICES ON LV DOORS LABELS COLOR	Black letters/white background
DEVICES INSIDE LV COMP. LABELS LANGUAGE	English
DEVICES INSIDE LV COMP. LABELS DESIGN	Self-adhesive Brother tape
DEVICES INSIDE LV COMP. LABELS COLOR	Black letters/white background
WARNING LABELS LANGUAGE	English
WARNING LABELS DESIGN	Glue
OTHER PARAMETERS	
SHUTTERS PAINTING	Standard RAL7035

FINAL DOCUMENTATION

NUMBER OF HARDCOPIES/CD DISC/FTP	0/1/Yes
STRUCTURE	Per typical
FORMAT	Pdf
CTs, VTs ROUTINE TEST REPORTS LANGUAGE	Not required
CTs, VTs MAGNETIZATION CURVES REQUIRED	Yes
CTs, VTs CALIBRATION FOR TARIFF METERING REQ.	No
CTs, VTs CORRECTION CURVES REQUIRED	Yes
MEASUREMENT OF PARTIAL DISCHARGE	No

SAMPLE

**) BECAUSE OF TECHNICAL REASONS, SOME BARS CAN BE SILVERED ALSO IN CASE WHEN "NO" IS SELECTED.
***) BECAUSE OF TECHNICAL REASONS, SOME BARS CANNOT BE INSULATED ALSO IN CASE WHEN "YES" IS SELECTED.

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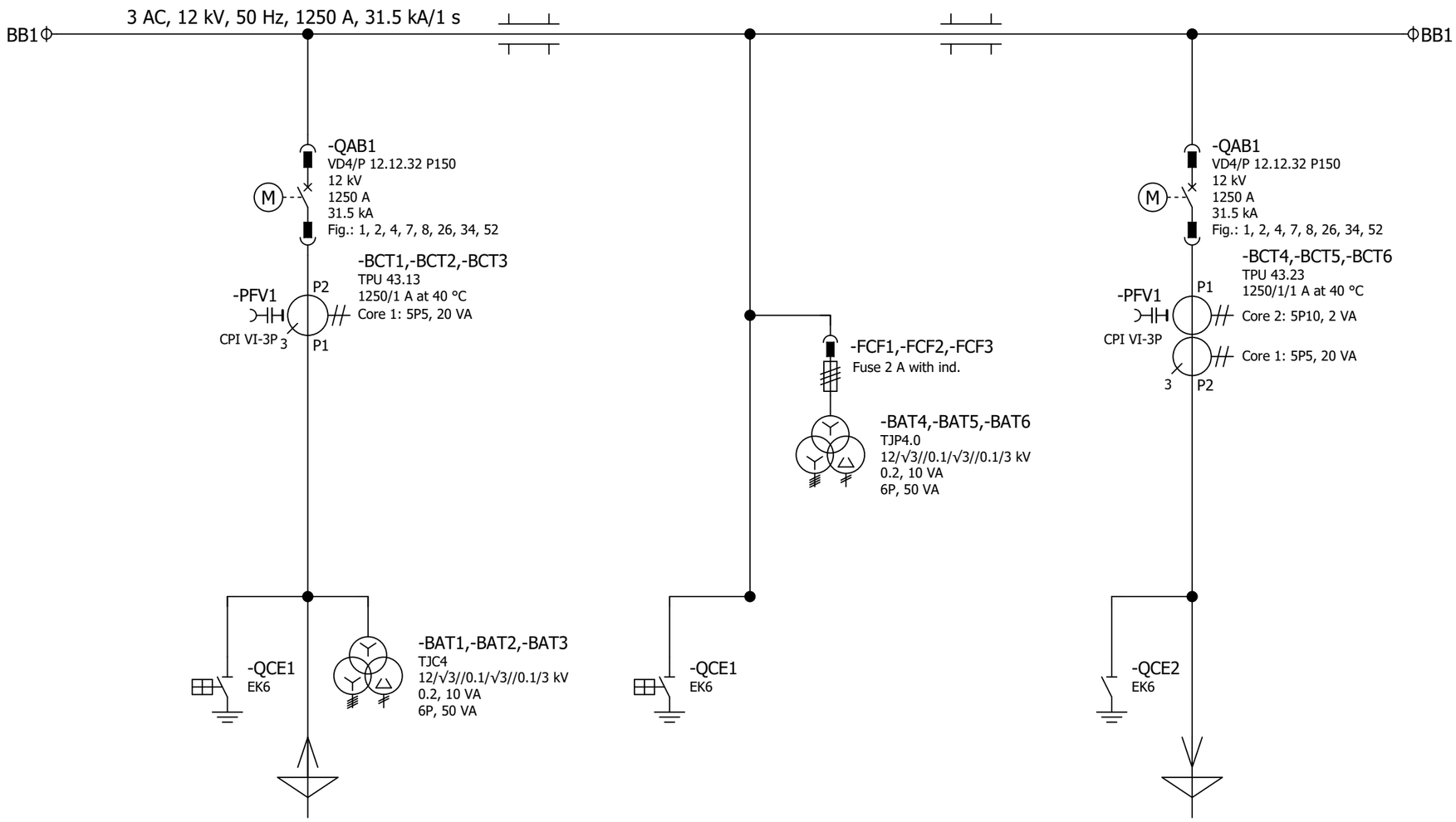
SAMPLE



PANEL	1	K01	2	K02	3	K03
PANEL NAME		Incoming Feeder		Metering		Outgoing Feeder
DRAWING Nr.		1VL84000001		1VL84000002		1VL84000003
PROT. RELAY		RELAY REF615 HBFEAEAGNBA1BNA11G		NONE		RELAY REF615 HBFDADAFNBA1BNN11G
PROT. FUNCT.		50P/51P, 67-1, 67-2, 67N-1, 67N-2, 27, 59, 59G		NONE		50P/51P, 67-1, 67-2, 67N-1, 67N-2, 27, 59, 59G
METERING FUNCT.		3I, 3V, Vn, P, E, f, DFR		NONE		3I, 3V, Vn, P, E, f, DFR
COMMUNICATION		NONE		NONE		NONE
METERING, TEST AND CONTROL EQUIPMENT		⊗⊗⊗				⊗⊗⊗

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K00
name	name	name			Customer document number			Place of installation +	SLD
Revision / Date	Name	Remark	Document number	1VL84000000	Customer	Page description	COMMON DRAWINGS SINGLE LINE DIAGRAM	DCC &	&EFA
			Date	date				Page	D01
			ABB s.r.o. EPDS Brno					Pages	2

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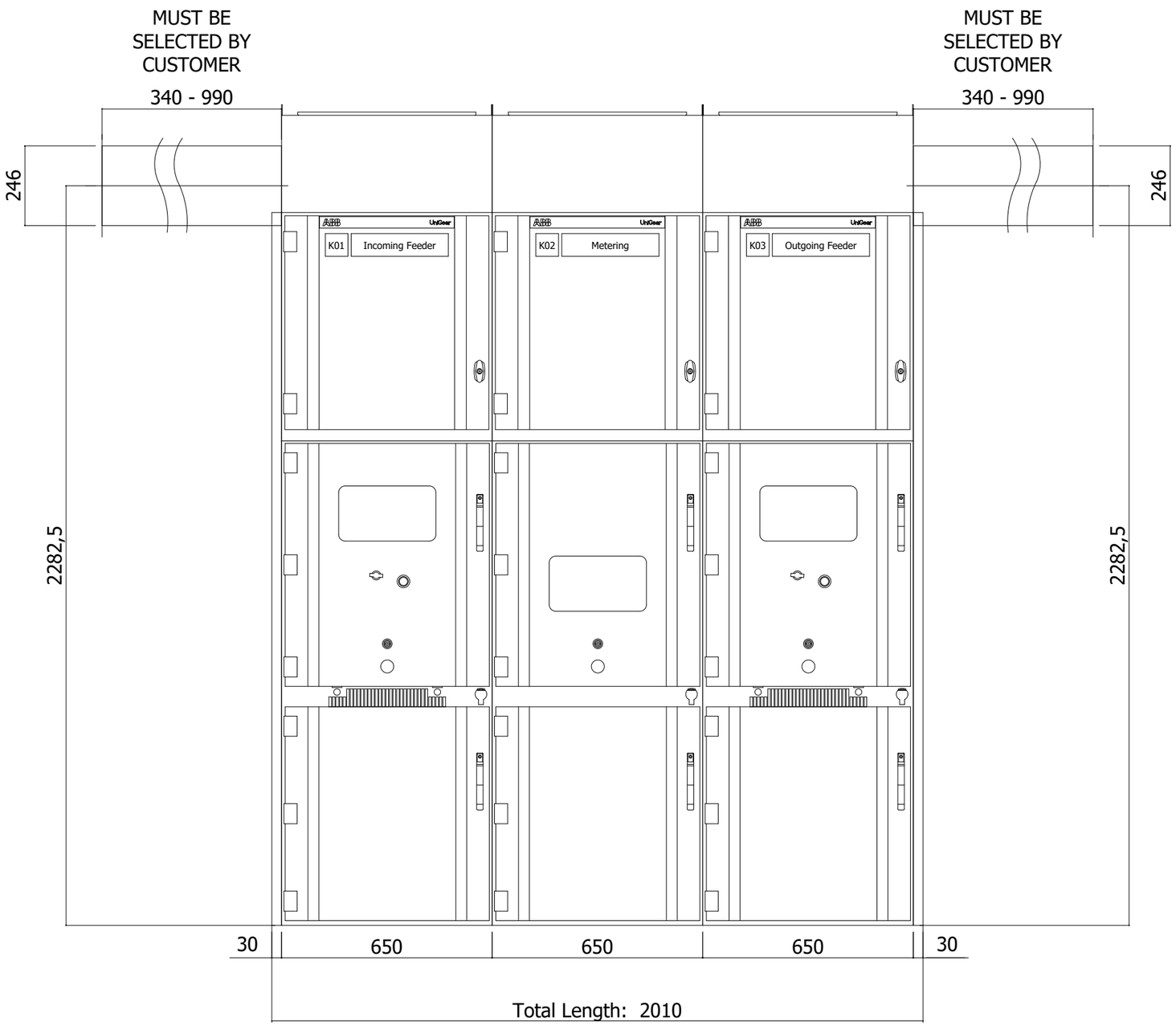
SAMPLE

Minimum switchgear room height - H: 2700 mm



PANEL	1	K01	2	K02	3	K03
PANEL NAME	Incoming Feeder		Metering		Outgoing Feeder	
CB MECHANICAL CONTROL	Off		No		Off	

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K00
name	name	name			Customer document number			Place of installation +	GA
Revision / Date / Name / Remark			Document number	1VL84000000	Customer	Page description	COMMON DRAWINGS FRONT VIEW	DCC &	&ELH
			Date	date				Page	F01
			ABB s.r.o. EPDS Brno					Pages	4

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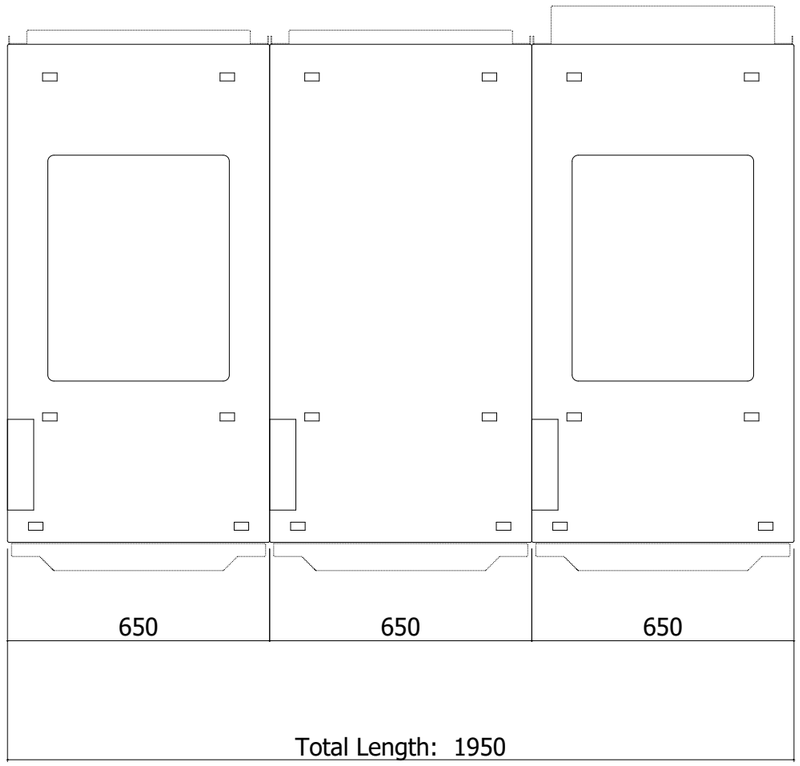
SAMPLE

Minimal distance between a wall and a panel must be for wall installation 100 mm (recommended 500 mm) lateral and on back side.



PANEL	1 K01	2 K02	3 K03
PANEL NAME	Incoming Feeder	Metering	Outgoing Feeder
MV CABLE	2x3x(1Cx630mm²)	-	2x3x(1Cx630mm²)
MV CABLEHOLDER SIZE	35 - 54 mm		35 - 54 mm
INTERLOCKING BETWEEN CABLE COMP. DOOR AND ES	Mechanical interlocking	Mechanical interlocking	Mechanical interlocking
INTERLOCKING BETWEEN CB DOOR AND WITHDRAW. PART	Mechanical interlocking	None	Mechanical interlocking
LOCK ON COVER FOR THE CB-RACK.-CRANK-INSERT. HOLE	Padlock	No	Padlock
LOCK ON COVER FOR THE ES-OPER.-LEVER-INSERT. HOLE	Padlock	Padlock	Padlock

SAMPLE

Drawn by	Checked by	Released by
name	name	name

Revision	Date	Name	Remark

Project number: 1098888888
Document number: 1VL84000000
Date: date

ABB s.r.o.
EPDS Brno

Customer project number
Customer document number
Customer

Project name: 12 kV Switchgear Sample
Page description: COMMON DRAWINGS SWGR. FOOTPRINT

Installation = K00
Place of installation + GA
DCC & &ELH
Page H01
Pages 4

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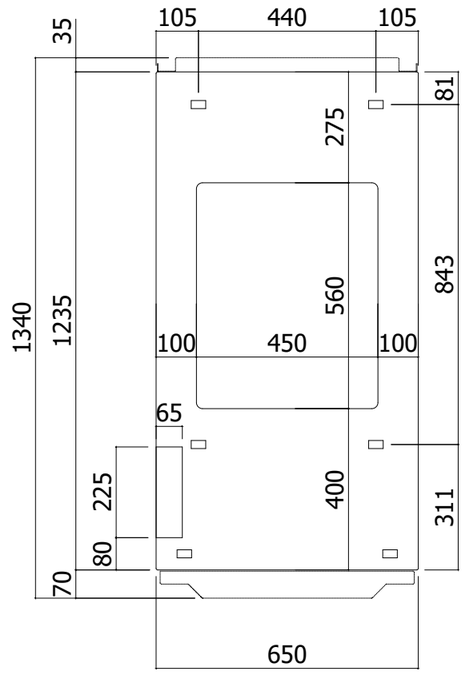
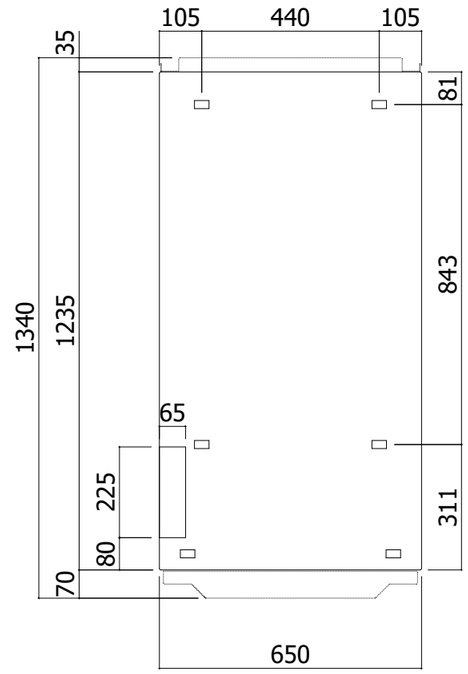
SAMPLE

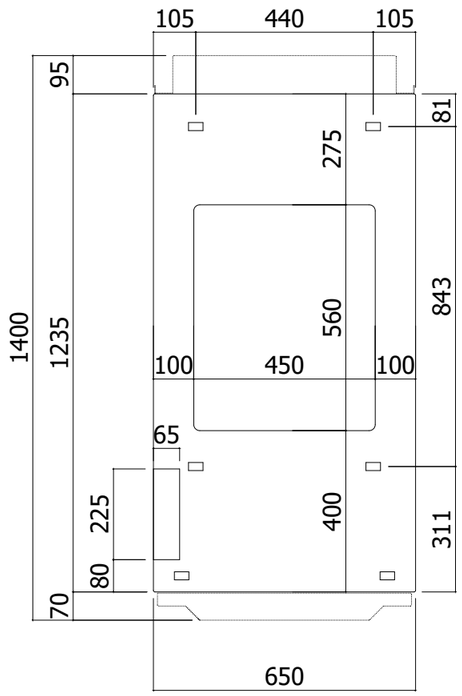
Tolerances for laying the floor frame are:
Evenness tolerance: ± 1 mm within a measuring length of 1 m
Straightness tolerance: 1 mm per 1 m, but not more than 3 mm over entire length of frame.
For more details refer to Installation, service and maintenance manual.



VALID FOR:
K01



VALID FOR:
K02



VALID FOR:
K03

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K00
name	name	name			Customer document number			Place of installation +	GA
Revision / Date	Name	Remark	Document number	1VL84000000	Customer	Page description	COMMON DRAWINGS SWGR. FOOTPRINT DETAILS	DCC &	&ELH
			Date	date				Page	J01
			ABB	ABB s.r.o. EPDS Brno				Pages	4

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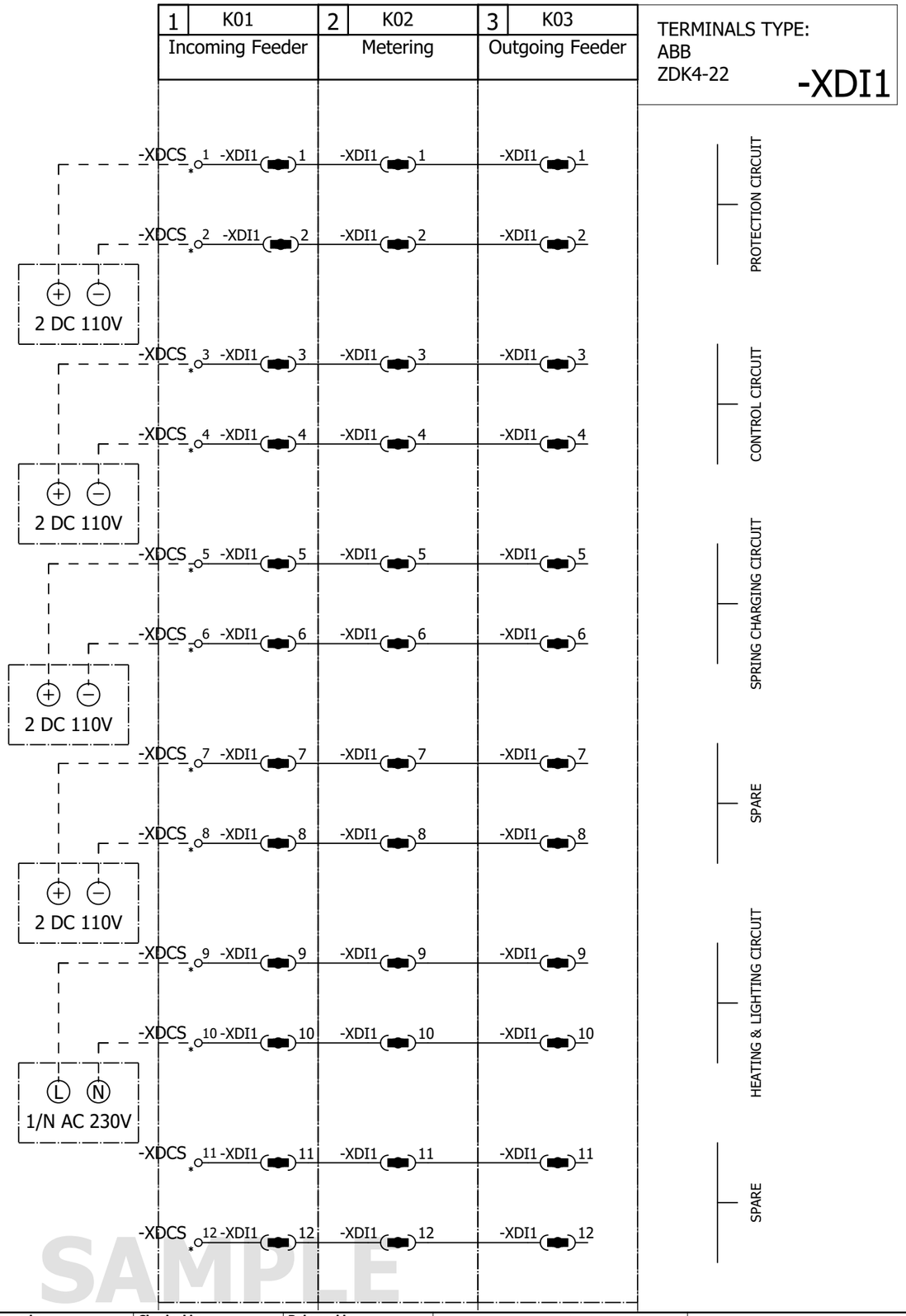
SAMPLE



Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K00
name	name	name			Customer document number			Place of installation +	ICL
Revision / Date / Name / Remark			Document number	1VL84000000	Customer	Page description	COMMON DRAWINGS XDI1 - SUPPLY INTERCONNECTION	DCC &	&EFF
			Date	date				Page	Q01
			ABB s.r.o. EPDS Brno					Pages	2

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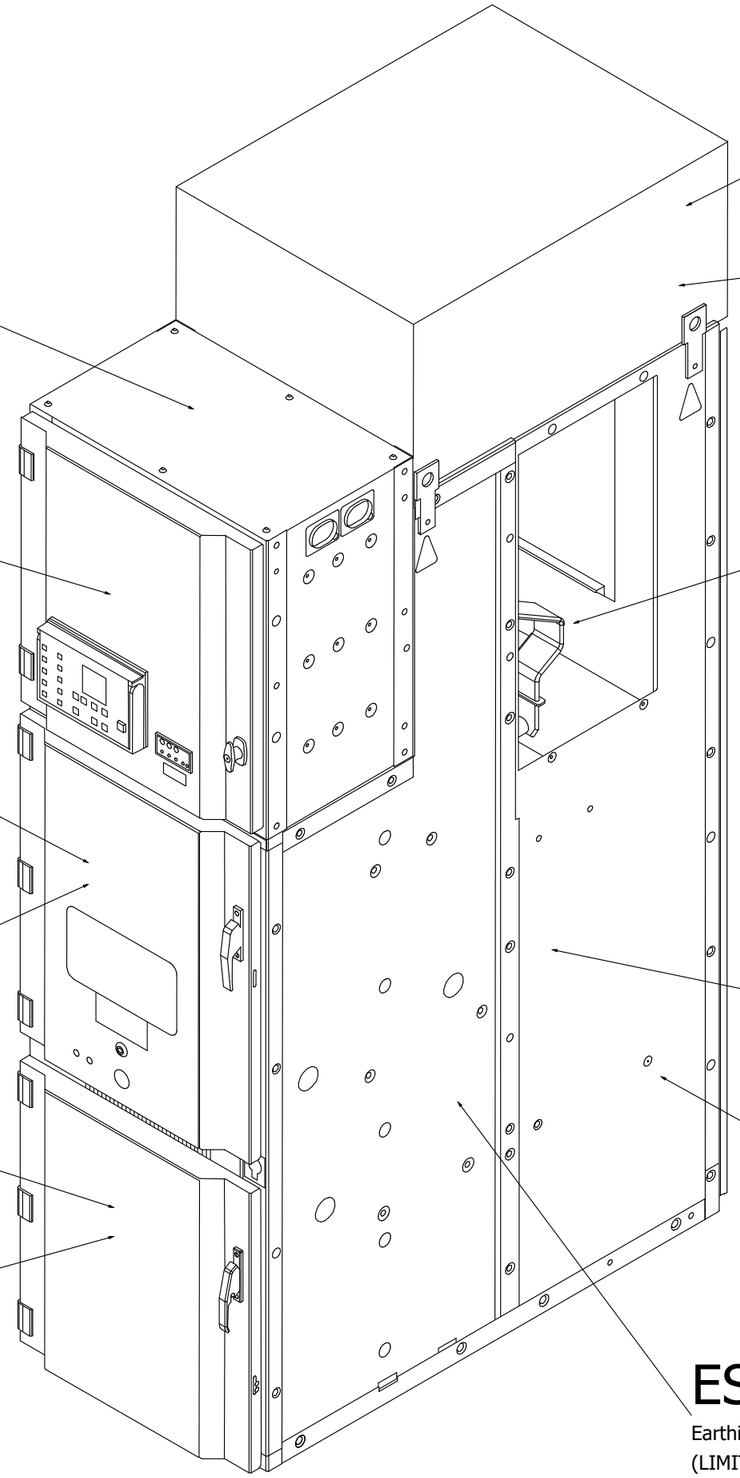
SAMPLE



TOP
TOP of switchgear
(ANTIFERRORESONANCE RESISTOR)

LVD
Low Voltage compartment Door
(DEVICES ON DOOR OF LV COMPARTMENT)

CB
Circuit Breaker compartment
(HEATER AND/OR ARC SENSOR IN BREAKER COMPARTMENT, WITHDRAWABLE VT METERING, ITH LIMITOR IN CB COMPARTMENT)

MV
Medium Voltage compartment
(LOAD BREAK SWITCH - NALFE)

CAB
CABle compartment
(SURGE ARRESTERS, GFCT*,ARC SENSOR, ITH LIMITOR IN CABLE COMPARTMENT)

CRT/CWT
Cable Removable/Withdrawable voltage Transformers
(REMOVABLE/WITHDRAWABLE VT IN CABLE COMPARTMENT)

TVT
Top mounted busbar Voltage Transformers
(TOP MOUNTED VOLTAGE TRANSFORMER, CAPACITIVE DIVIDERS, C.D. CABLES)

TES
Top mounted busbar Earthing Switch
(TOP MOUNTED EARTHING SWITCH, CAPACITIVE DIVIDERS, C.D. CABLES)

BB
BusBar compartment
(ARC DETECTOR ON BUSBAR, ITH LIMITOR IN BB COMPARTMENT)

CT
Current Transformer
(CURRENT TRANSFORMERS, CAPACITIVE DIVIDERS, C.D. CABLES)

VT
Voltage Transformer
(FIX MOUNTED VOLTAGE TRANSFORMERS)

ES
Earthing Switch
(LIMIT SWITCHES AND COIL OF EARTHING SWITCH)

SAMPLE

* - GFCT (Ground Fault Current Transformer) also in case if are mounted under cubicle. For place of mounting see SLD diagram of switchgear

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K00
name	name	name			Customer document number				
Revision / Date	Name	Remark	Document number	1VL84000000	Customer			Place of installation +	INF
			Date	date		Page description	COMMON DRAWINGS ZONES IN EPLAN - MV	DCC &	&EFS
			ABB	ABB s.r.o. EPDS Brno				Page	Z01
								Pages	12

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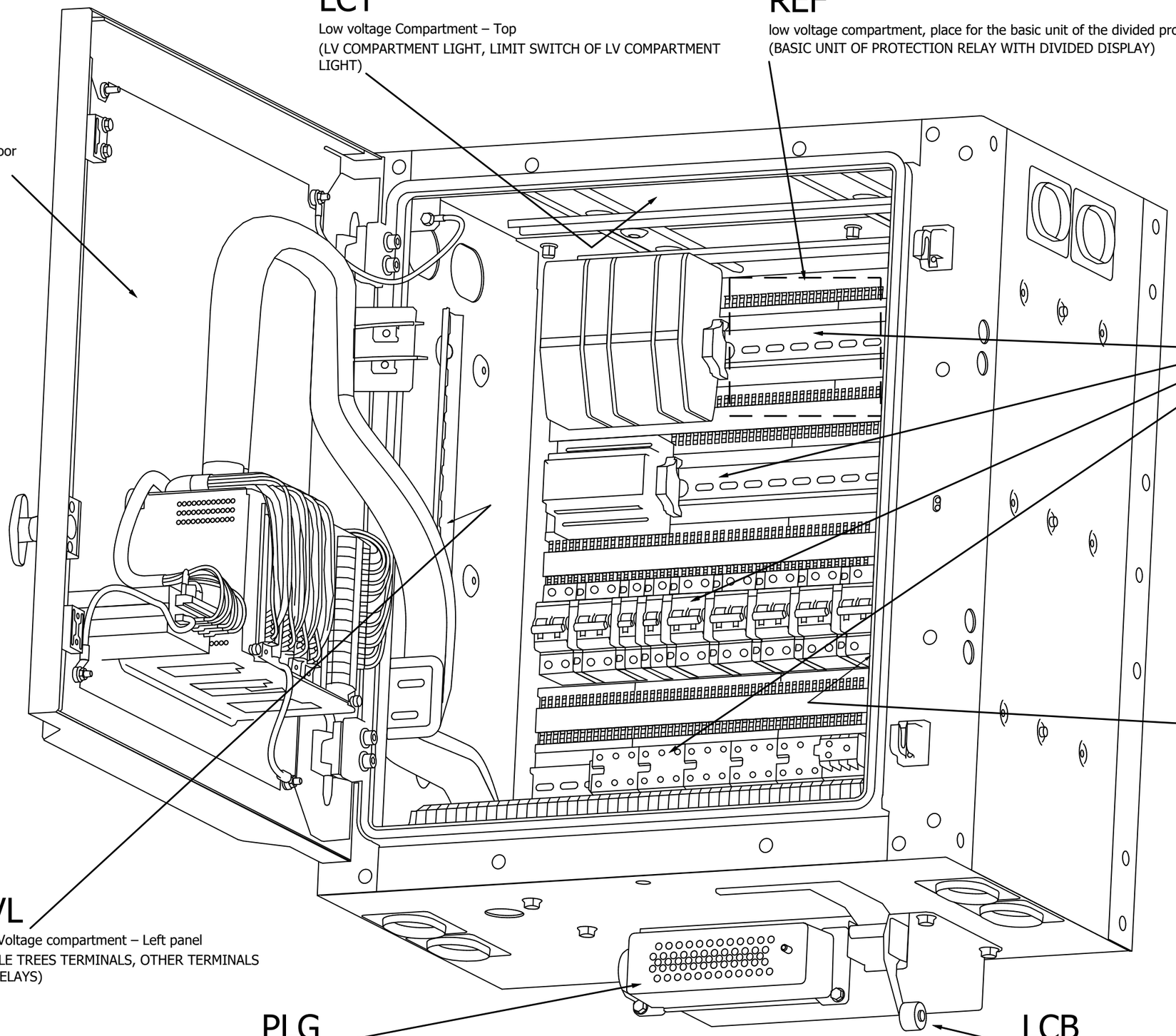
SAMPLE



LCT
Low voltage Compartment – Top
(LV COMPARTMENT LIGHT, LIMIT SWITCH OF LV COMPARTMENT LIGHT)

REF
low voltage compartment, place for the basic unit of the divided protection device
(BASIC UNIT OF PROTECTION RELAY WITH DIVIDED DISPLAY)

LVD
Low Voltage compartment Door
(LV DOOR COMPARTMENT)

LV
wiring panel of Low Voltage compartment
(DEVICES AND TERMINALS INSIDE LV COMPARTMENT)

LVR + LCR
Low Voltage compartment – Right panel
(CABLE TREES TERMINALS, LV HEATER + LOW VOLTAGE COMPARTMENT LIGHT)

LVL
Low Voltage compartment – Left panel
(CABLE TREES TERMINALS, OTHER TERMINALS OR RELAYS)

PLG
medium voltage compartment – truck multiPLuG
(TRUCK MULTIPLUG)

LCB
Low voltage Compartment – Bottom
(LIMIT SWITCH OF TRUCK MULTIPLUG)

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K00
name	name	name			Customer document number				
Revision / Date	Name	Remark	Document number	1VL84000000	Customer	Page description	COMMON DRAWINGS ZONES IN EPLAN - LV	Place of installation +	INF
			Date	date				DCC & &EFS / Page Z02 / Pages 12	
			ABB s.r.o. EPDS Brno						

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SAMPLE

DESCRIPTION OF ZONES ASSIGNED IN EPLAN

ZONE	DESCRIPTION
ACS	ACCESSORIES
BB	BUSBAR COMPARTMENT
CAB	CABLE COMPARTMENT
CB	CIRCUIT BREAKER COMPARTMENT
CRT (CWT)	CABLE REMOVABLE (WITHDRAWABLE) VOLTAGE TRANSFORMERS
CT	CURRENT TRANSFORMERS
ES	EARTHING SWITCH
LCB	LOW VOLTAGE COMPARTMENT-BOTTOM
LCR	LOW VOLTAGE COMPARTMENT-RIGHT PANEL
LCT	LOW VOLTAGE COMPARTMENT-TOP
LV	WIRING PANEL OF LOW VOLTAGE COMPARTMENT
LVD	LOW VOLTAGE COMPARTMENT DOOR
LVR	WIRING PANEL OF LOW VOLTAGE COMPARTMENT-RIGHT PANEL
MV	MEDIUM VOLTAGE COMPARTMENT
PLG	MEDIUM VOLTAGE COMPARTMENT-TRUCK MULTIPLUG
REF	LOW VOLTAGE COMPARTMENT- BASIC UNIT OF DIVIDED REF PROTECTION
SPA	PROJECT SPARE PARTS
TES	TOP MOUNTED BUSBAR EARTHING SWITCH
TOP	TOP OF SWITCHGEAR (FOR DUMPING RESISTOR)
TRA	TRANSFORMER ACCESSORIES
TVT	TOP MOUNTED BUSBAR VOLTAGE TRANSFORMERS
VT	VOLTAGE TRANSFORMERS

SAMPLE

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SAMPLE

DEVICE NAME DESCRIPTION

DEVICE NAME	DESCRIPTION
-AA	MULTIFUNCTION UNIT (CENTRAL UNITS)
-BAD	CAPACITIVE VOLTAGE DIVIDERS
-BAR	VOLTAGE PROTECTION RELAYS
-BAS	VOLTAGE SENSORS
-BAT	VOLTAGE TRANSFORMERS
-BCD	DIFFERENTIAL PROTECTION RELAYS
-BCF	FEEDER PROTECTION RELAYS
-BCG	GENERATOR PROTECTION RELAYS
-BCM	MOTOR PROTECTION RELAYS
-BCN	NEUTRAL CURRENT TRANSFORMERS
-BCP	TRANSFORMER PROTECTION RELAYS
-BCR	CURRENT PROTECTION RELAYS
-BCS	CURRENT SENSORS
-BCT	CURRENT TRANSFORMERS
-BCZ	DISTANCE PROTECTION RELAYS
-BD	DENSITY SWITCHES
-BEF	FREQUENCY SCHUTZ RELAYS
-BER	SUPERVISONS RELAYS
-BES	SYNCHRONIZING RELAYS
-BET	THERMAL PROTECTION RELAYS
-BGA	ARC SCHOCK DETECTORS (POSITION SWITCHES)
-BGB	POSITION SWITCHES (PROXIMITY SENSORS) OF CIRCUIT-BREAKER
-BGC	POSITION SWITCHES OF CIRCUIT-BREAKER CONNECTOR
-BGD	POSITION SWITCHES (PROXIMITY SENSORS) OF THE DOOR
-BGE	POSITION SWITCHES (PROXIMITY SENSORS) OF EARTHING SWITCH

DEVICE NAME	DESCRIPTION
-BGF	POSITION SWITCHES OF MEDIUM VOLTAGE FUSES
-BGI	POSITION SWITCHES OF DISCONNECTORS
-BGK	POSITION SWITCHES OF KEY LOCKS
-BGL	POSITION SWITCHES OF ELECTROMECHANICAL LOCKS
-BGS	POSITION SWITCHES (PROXIMITY SENSORS) OF CIRCUIT-BREAKER SPRINGS
-BGT	POSITION SWITCHES (PROXIMITY SENSORS) OF CIRCUIT-BREAKER TRUCK
-BGV	POSITION SWITCHES OF VENTILATORS
-BGZ	OTHER POSITION SWITCHES
-BM	HYGROSTAT
-BPA	ARC PRESSURE DETECTORS
-BPS	PRESSURE SWITCHES
-BQ	INSULATION MONITORING DEVICE
-BR	FLAME DETECROS, SMOKE DETECTORS, OPTOCOUPLER
-BT	THERMOSTATS
-BU	BUCHHOLZ RELAY
-BX	ARC PROTECTION RELAYS
-BZ	MULTIFUNCTION SENSORS, COMBINED CURRENT AND VOLTAGE SENSORS
-BZR	RADIO FREQUENCY IDENTIFICATION DEVICES
-CA	CAPACITORS
-CN	GAS CYLINDER
-CQ	SPRING
-EA	LIGHTING LAMPS
-EB	HEATERS
-FA	SURGE ARRESTER
-FB	RESIDUAL CURRENT DEVICE

SAMPLE

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SAMPLE

DEVICE NAME DESCRIPTION

DEVICE NAME	DESCRIPTION
-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS
-FCF	MEDIUM VOLTAGE FUSES
-FCM	MINIATURE CIRCUIT-BREAKERS
-FCT	THERMAL OVERLOAD RELEASES
-FN	MECHANICAL OR INSULATING PROTECTION
-GA	GENERATORS
-GQ	VENTILATORS
-KFA	AUXILIARY RELAYS OR CONTACTORS
-KFC	AUXILIARY RELAYS OR CONTACTORS FOR CLOSING
-KFE	INPUT/OUTPUT MODULE, INPUT/OUTPUT EXPANSION MODULE
-KFI	INTEGRATED CIRCUITS
-KFL	LOCK-OUT RELAYS
-KFM	MICROPROCESSORS
-KFN	ANTIPUMPING RELAYS
-KFO	AUXILIARY RELAYS OR CONTACTORS FOR OPENING
-KFP	PROGRAMMABLE LOGIC CONTROLLER (PLC)
-KFR	RECLOSING RELAYS
-KFS	SYNCHRONIZING DEVICES
-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS
-KFU	CONTROL UNITS
-KFZ	FEEDBACK CONTROLLER, POTENTIOMETER, REGULATOR
-MAD	MOTORS FOR ELECTRICAL OPERATION OF DISCONNECTORS
-MAE	MOTORS FOR ELECTRICAL OPERATION OF EARTHING SWITCH
-MAS	MOTORS FOR SPRINGS CHARGING OF CIRCUIT-BREAKER OR DISCONNECTOR
-MAT	MOTORS FOR ELECTRICAL OPERATION OF TRUCK

DEVICE NAME	DESCRIPTION
-MBC	CLOSING RELEASES
-MBO	OPENING RELEASE
-MBU	UNDERVOLTAGE RELEASES
-ML	OPERATING MECHANISM
-PFB	BLUE SIGNAL LAMPS
-PFF	FLAG RELAYS
-PFG	GREEN SIGNAL LAMPS
-PFR	RED SIGNAL LAMPS
-PFS	SHORT CIRCUIT INDICATORS
-PFV	VOLTAGE INDICATORS
-PFW	WHITE SIGNAL LAMPS
-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS
-PFY	YELLOW SIGNAL LAMPS
-PGA	AMMETERS
-PGC	COUNTERS
-PGF	FREQUENCY METERS
-PGH	HOUR METERS
-PGI	HUMAN MACHINE INTERFACES
-PGJ	ACTIVE ENERGY METERS
-PGK	REACTIVE ENERGY METERS
-PGM	MULTIFUNCTION INDICATORS
-PGP	POWER-FACTOR METERS
-PGQ	VARMETERS
-PGS	SYNCHRONOSCOPES
-PGV	VOLTMETERS

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SAMPLE

DEVICE NAME DESCRIPTION

DEVICE NAME	DESCRIPTION
-PGW	WATTMETERS
-PH	NAMEPLATES
-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)
-QAB	CIRCUIT-BREAKERS
-QAC	CONTACTORS (FOR POWER)
-QBD	DISCONNECTORS
-QBH	MANUAL CIRCUIT-BREAKERS
-QBL	LINKS
-QBM	MINIATURE SWITCH-DISCONNECTORS
-QBP	POLES OR VACUUM INTERRUPTER
-QBS	SWITCH-DISCONNECTORS
-QBT	TRUCKS
-QCA	EARTHING SWITCHES FOR ELIMINATION OF INTERNAL ARC
-QCE	EARTHING SWITCHES
-QQ	CLUTCHES
-QZD	THREE-POSITION DISCONNECTORS (COMBINED DISCONNECTOR AND EARTHING SWITCHES)
-QZS	THREE-POSITION SWITCH-DISCONNECTORS (COMBINED SWITCH-DISCONNECTORS AND EARTHING SWITCHES)
-RAA	ANTIFERRORESONANCE RESISTORS
-RAD	DIODES
-RAI	INDUCTORS
-RAR	RESISTORS
-RF	FILTERS
-RLE	ELECTROMECHANICAL LOCKS
-RLM	MECHANICAL INTERLOCKS
-RR	SHOCK-ABSORBERS

DEVICE NAME	DESCRIPTION
-SFA	AMMETRIC SWITCHES
-SFC	CLOSING PUSH-BUTTONS, CONTROL SWITCHES
-SFL	LOCKING CONTACTS
-SFM	MOTOR CONTROL PUSH-BUTTON
-SFO	OPENING PUSH-BUTTONS
-SFR	RESET PUSH-BUTTONS
-SFS	SELECTOR SWITCHES
-SFT	TEST PUSH-BUTTONS
-SFU	UNLOCKING PUSH-BUTTONS
-SFV	VOLTMETRIC SWITCHES
-TA	POWER TRANSFORMERS
-TB	RECTIFIERS, CONVERTERS (FEEDERS)
-TFA	ACTIVE POWER TRANSDUCERS
-TFC	CURRENT TRANSDUCERS
-TFF	FREQUENCY TRANSDUCERS
-TFJ	ACTIVE ENERGY TRANSDUCERS
-TFK	REACTIVE ENERGY TRANSDUCERS
-TFM	MULTIFUNCTION TRANSDUCERS
-TFP	POWER-FACTOR TRANSDUCERS
-TFQ	REACTIVE POWER TRANSDUCERS
-TFS	SIGNAL CONVERTERS
-TFT	TEMPERTAURE TRANSDUCERS
-TFV	VOLTAGE TRANSDUCERS
-UA	SUPPORTS
-UZ	FITTINGS

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DEVICE NAME DESCRIPTION

DEVICE NAME	DESCRIPTION
-WA	BUSBARS
-WBB	BUSHINGS
-WBC	POWER CABLES
-WE	EARTHING CONNECTIONS
-WGC	CONTROL CABLES
-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)
-WH	OPTICAL FIBRES
-WQ	MECHANICAL TRANSMISSIONS
-XB	POWER CONNECTORS
-XD	CONNECTORS OR TERMINAL BLOCKS FOR MEDIUM VOLTAGE SWITCHES (ADMITTED)
-XDA	TERMINAL BLOCKS FOR AMMETRIC CIRCUITS
-XDB	CONNECTORS OR TERMINAL BLOCKS FOR CIRCUIT-BREAKER
-XDC	TERMINAL BLOCKS FOR CUSTOMER
-XDE	CONNECTORS OR TERMINAL BLOCKS FOR EARTHING SWITCH
-XDF	CONNECTORS OR TERMINAL BLOCKS FOR FORCED COOLING VENTILATORS
-XDH	CONNECTORS OR TERMINAL BLOCKS FOR HEATERS AND FOR POSITION SWITCHES DETECTING INTERNAL ARC
-XDI	CONNECTORS OR TERMINAL BLOCKS FOR INTERCONNECTIONS
-XDJ	CONNECTORS OR TERMINAL BLOCKS FOR DISCONNECTORS
-XDM	TERMINAL BLOCKS FOR MEASURING TEST
-XDP	MULTILEVEL TERMINAL BLOCKS WITH ADVANCED BRIDGING OPTIONS
-XDS	SOCKET OUTLETS
-XDT	CONNECTORS OR TERMINAL BLOCKS FOR TRUCK POSITION CONTACTS
-XDU	CONNECTORS OR TERMINAL BLOCKS FOR VOLTAGE TRANSFORMER TRUCKS
-XDV	CONNECTORS OR TERMINAL BLOCKS FOR VOLTMETRIC CIRCUITS
-XDX	SUPPORT TERMINAL BLOCKS OR SUPPORT CONNECTORS
-XDZ	CONNECTORS OR TERMINAL BLOCKS FOR THREE-POSITION DISCONNECTORS OR SWITCH-DISCONNECTORS
-XE	EARTHING TERMINALS

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TERMINAL STRIP DESCRIPTION

TERMINAL STRIP	DESCRIPTION
-XDI1	INTERCONNECTIONS
-XDI2	INTERCONNECTIONS
-XDI3	INTERCONNECTIONS
-XDI4	INTERCONNECTIONS
-XDIV	INTERCONNECTIONS
-XDIA	INTERCONNECTIONS
-XDA	CURRENT TRANSFORMERS
-XDV	VOLTAGE TRANSFORMER
-XDCA	CURRENTS TO CUSTOMER SIDE
-XDCV	VOLTAGES TO CUSTOMER SIDE
-XDB1	TERMINAL BLOCKS FOR CB AUXILIARY CIRCUITS
-XDB2	END SWITCH OF CB 58-PIN PLUG
-XDJ	TERMINAL BLOCK FOR DISCONNECTOR AUXILIARY CIRCUIT
-XDX	CONTROL AND PROTECTION CIRCUITS
-XDX1	PROTECTION RELAY CARDS
-XDH	AC 230 V CIRCUITS
-XE	GROUND
-XDE	EARTHING SWITCH
-XDF	TERMINALS BLOCK FOR AUXILIARY CIRCUIT OF FORCED COOLING VENTILATORS
-XDH	ITH LIMITORS
-XDC	CUSTOMER TERMINALS
-XDCS	AUXILIARY SUPPLY TERMINALS
-WE	EARTHING POINTS

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SAMPLE

PROTECTION FUNCTIONS - ANSI CODES

FUNCTION	DESCRIPTION
21, 21P, 21N	DISTANCE PROTECTION
21FL	FAULT LOCATOR
21GT	THREE-PHASE UNDERIMPEDANCE PROTECTION
21YN	ADMITTANCE BASED EARTH-FAULT PROTECTION
24	OVEREXCITATION PROTECTION
27	THREE-PHASE UNDERVOLTAGE PROTECTION
27/59THD	THIRD HARMONIC BASED STATOR EARTH-FAULT PROTECTION
32N	WATTMETRIC BYSED EARTH-FAULT PROTECTION
32R/32O	REVERSE POWER/DIRECTIONAL OVERPOWER PROTECTION
32U	UNDERPOWERPROTECTION
37F	LOSS OF PHASE (UNDERCURRENT)
37, 37M	LOSS OF LOAD SUPERVISION
40	THREE-PHASE UNDEREXCITATION PROTECTION
46	NEGATIVE-SEQUENCE OVERCURRENT PROTECTION
46G, 46M	NEGATIVE-SEQUENCE OVERCURRENT PROTECTION (FOR MOTORS, GENERATORS)
46PD	PHASE DISCONTINUITY PROTECTION
46R	PHASE REVERSAL PROTECTION
47O+	POSITIVE-SEQUENCE OVERVOLTAGE PROTECTION

FUNCTION	DESCRIPTION
47O-	NEGATIVE-SEQUENCE OVERVOLTAGE PROTECTION
47U+	POSITIVE-SEQUENCE UNDERVOLTAGE PROTECTION
49F	THREE-PHASE THERMAL PROTECTION FOR FEEDERS, CABLES AND DISTRIBUTION TRANSFORMERS
49T, 49G	THREE-PHASE THERMAL OVERLOAD PROTECTION (POWER TRANSFORMERS, GENERATORS)
49M	THERMAL OVERLOAD PROTECTION FOR MOTORS
50L/50NL	ARC PROTECTION
50N	NON-DIRECTIONAL EARTH-FAULT INSTANTANEOUS PROTECTION
50P	THREE-PHASE NON-DIRECTIONAL INSTANTANEOUS OVERCURRENT PROTECTION
51BF/51NBF	CIRCUIT BREAKER FAILURE PROTECTION
51LR	MOTOR LOAD JAM PROTECTION
51N	NON-DIRECTIONAL EARTH-FAULT PROTECTION
51NHA	HARMONIC BASED EARTH-FAULT PROTECTION
51P	THREE-PHASE NON-DIRECTIONAL OVERCURRENT PROTECTION
51V	VOLTAGE DEPENDENT OVERCURRENT PROTECTION
59	THREE-PHASE OVERVOLTAGE PROTECTION
59G	RESIDUAL OVERVOLTAGE PROTECTION
64	ROTOR EARTH-FAULT PROTECTION
67	THREE-PHASE DIRECTIONAL OVERCURRENT PROTECTION

NOTE: All protection functions mentioned on this page can be used in ABB Relion protection relays. Not all of protection functions are usable in every ABB Relion protection type.

SAMPLE

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SAMPLE

PROTECTION FUNCTIONS - ANSI CODES

FUNCTION	DESCRIPTION
67N	DIRECTIONAL EARTH-FAULT PROTECTION
67NIEF	TRANSIENT / INTERMITTENT EARTH-FAULT PROTECTION
68	THREE-PHASE INRUSH DETECTOR
81	FREQUENCY PROTECTION
81LSH	LOAD SHEDDING
81O	OVERFREQUENCY PROTECTION
81R	FREQUENCY GRADIENT PROTECTION
81U	UNDERFREQUENCY PROTECTION
87	HIGH-IMPEDANCE DIFFERENTIAL PROTECTION
87G/87M	STABILIZED DIFFERENTIAL PROTECTION FOR MACHINES
87GH/87MH	HIGH-IMPEDANCE OR FLUX-BALANCE BASED DIFFERENTIALPROTECTION FOR MACHINES
87M	MOTOR DIFFERENTIAL PROTECTION
87NH	HIGH IMPEDANCE BASED RESTRICTED EARTH-FAULT PROTECTION
87NL	NUMERICAL STABILIZED LOW IMPEDANCE RESTRICTED EARTH-FAULT PROTECTION
87L	LINE DIFFERENTIAL PROTECTION WITH IN ZONE POWER TRANSFORMER
87T	STABILIZED AND INSTANTANEOUS DIFFERENTIAL PROTECTION FOR 2-WINDING TRANSFORMERS
94/86	MASTER TRIP
94	TRIPPING LOGIC

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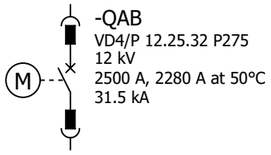
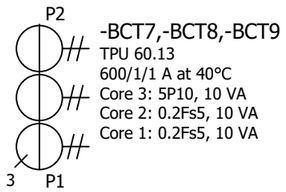
SAMPLE

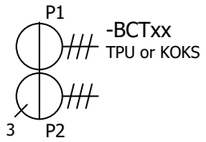
LEGEND AND SYMBOLS



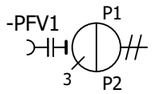
CIRCUIT BREAKER



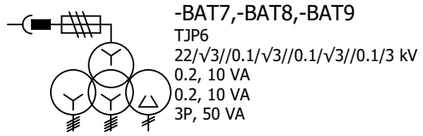
3 CORE CURRENT TRANSFORMER



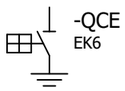
2 CORE SWITCHABLE CURRENT TRANSFORMER



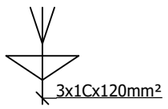
CORE OF CT WITH VOLTAGE PRESENCE INDICATOR



3 CORES VOLTAGE TRANSFORMER WITH FUSES

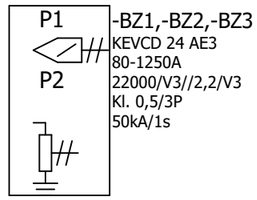


LINE EARTHING SWITCH EK6 WITH BLOCKING MAGNET

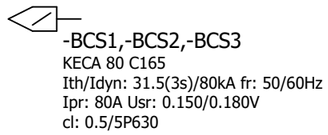


BUSHING

RING CORE CT KOLMA

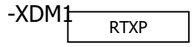


COMBINED SENSOR



CURRENT SENSOR

VOLTAGE PRESENCE INDICATOR



TEST BLOCK

A - METR

V - METR

WAT - METR

STATUS INDICATOR

PUSH BUTTON

SELECTOR SWITCH

TRANSDUCER

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K00
name	name	name			Customer document number				
Revision / Date	Name	Remark	Document number	1VL84000000	Customer			Place of installation +	INF
			Date	date		Page description	COMMON DRAWINGS LEGEND AND SYMBOLS	DCC & / Page / Pages	&EFS / Z15 / 12
			ABB	ABB s.r.o. EPDS Brno					

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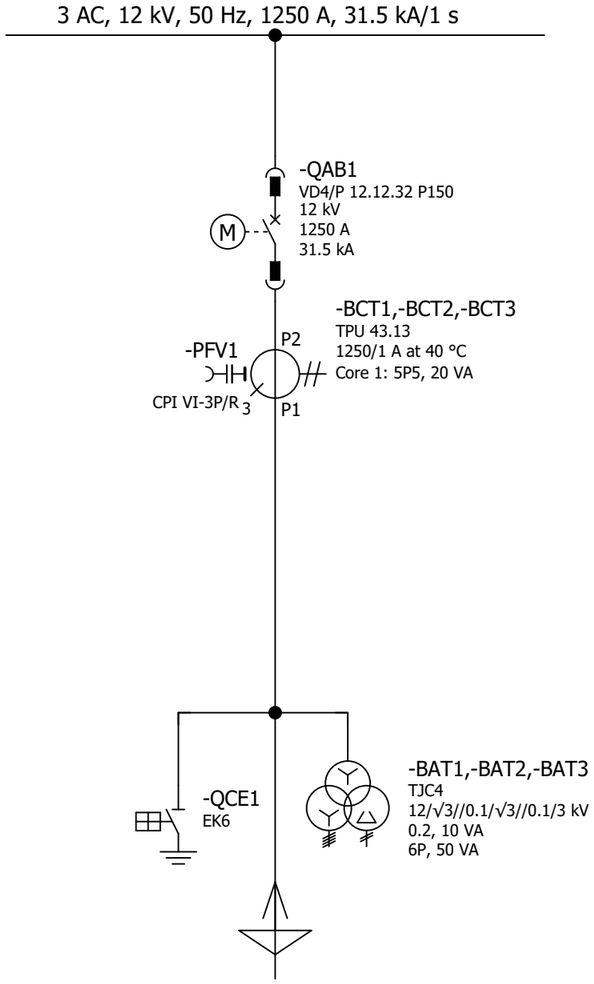
SAMPLE



VALID FOR PANELS:

POS.	DESIGN.	NAME
1	K01	Incoming Feeder

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K01
name	name	name			Customer document number			Place of installation +	LV
Revision / Date / Name / Remark			Document number	1VL84000001	Customer	Page description	Incoming Feeder SINGLE LINE SCHEME	DCC &	&EFA
			Date	date				Page	A01
			ABB s.r.o. EPDS Brno					Pages	28

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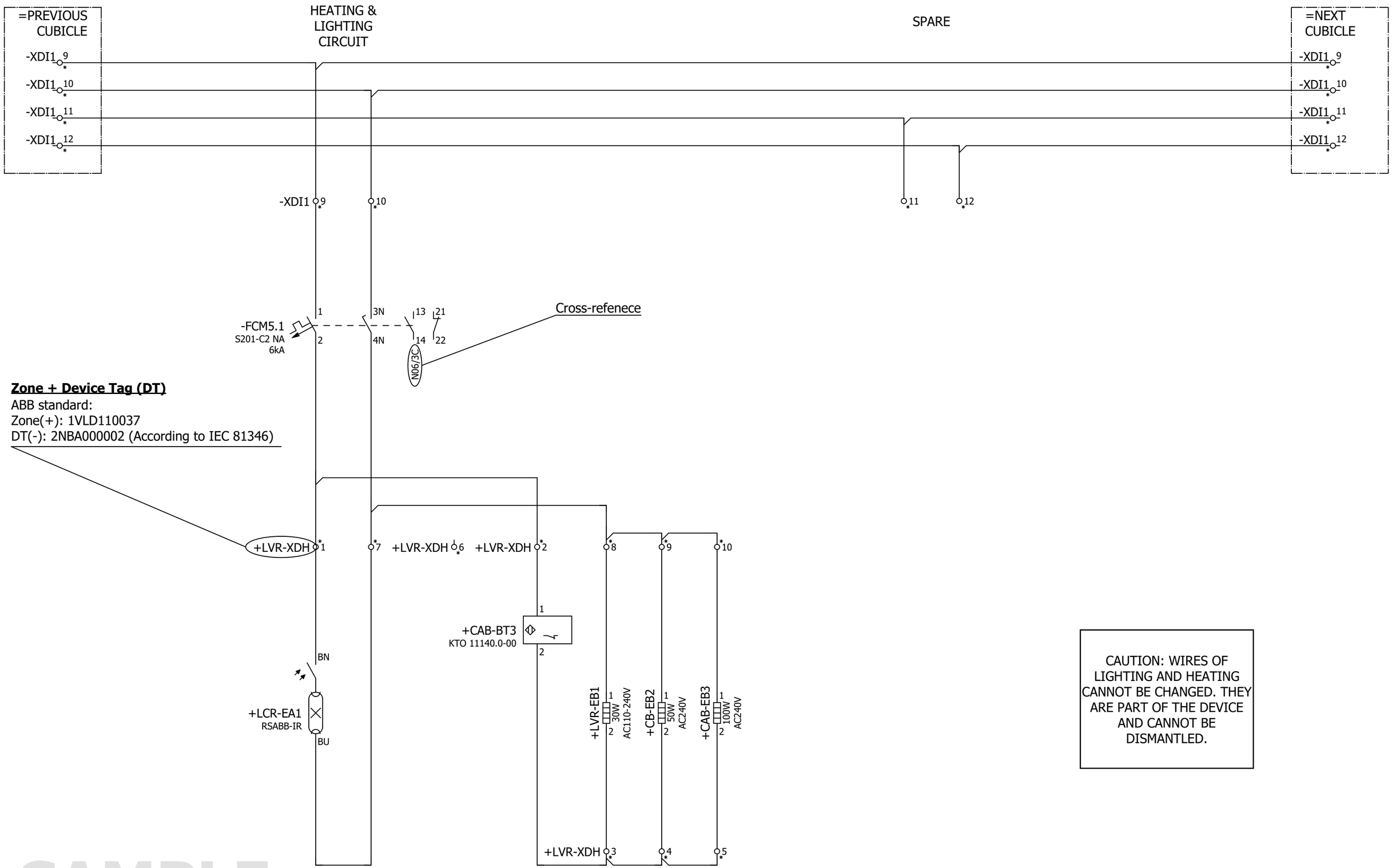
SAMPLE



Zone + Device Tag (DT)
ABB standard:
Zone(+): 1VLD110037
DT(-): 2NBA000002 (According to IEC 81346)

CAUTION: WIRES OF LIGHTING AND HEATING CANNOT BE CHANGED. THEY ARE PART OF THE DEVICE AND CANNOT BE DISMANTLED.

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K01
name	name	name			Customer document number				
Revision / Date	Name	Remark	Document number	1VL84000001	Customer	Page description	Incoming Feeder AC SUPPLY CIRCUITS	Place of installation +	LV
			Date	date				DCC &	&EFS
			ABB s.r.o. EPDS Brno					Page	D01
								Pages	28

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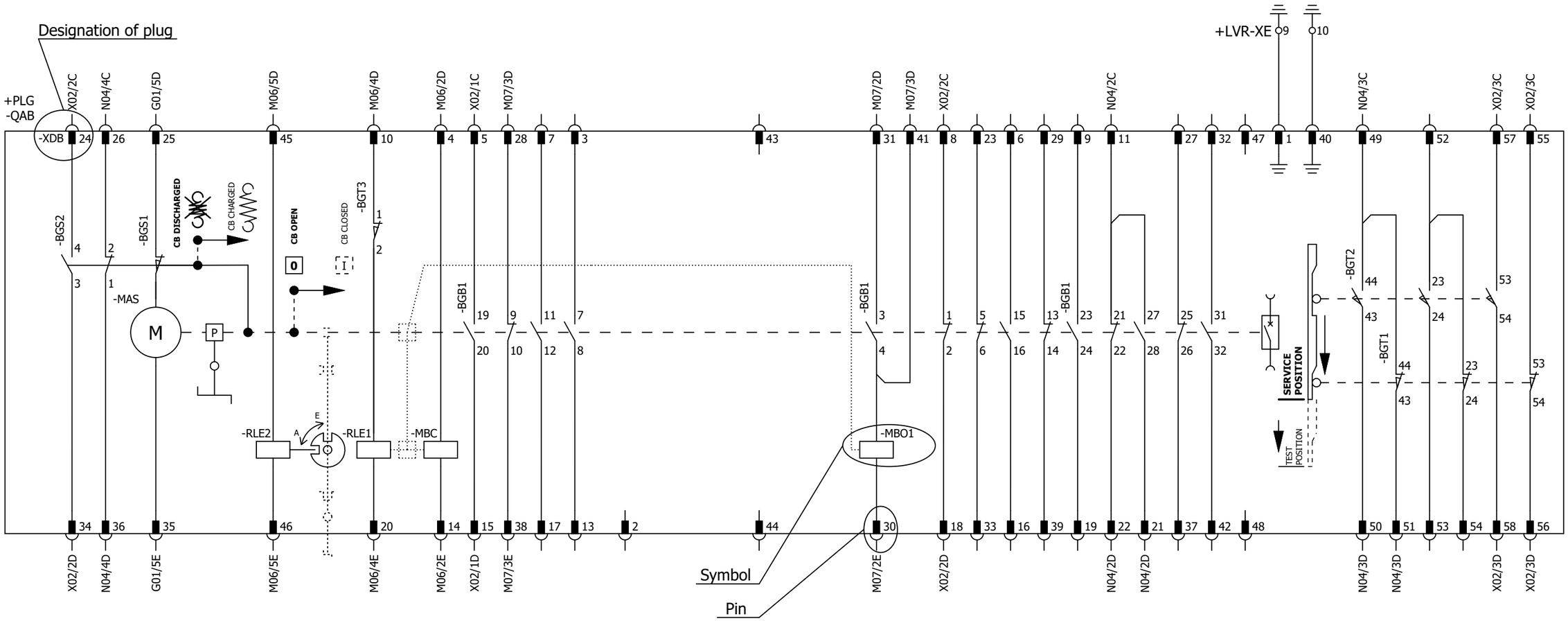
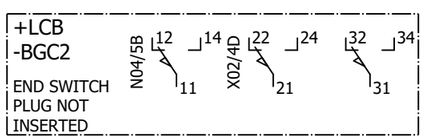
SAMPLE

VD4

DOC. NO: 1VCD400155

CB OPEN, TRUCK IN SERVICE POSITION, SPRING NOT CHARGED





Functional designation

DESIGNAT.	FIGURE	DESCRIPTION	DESIGNAT.	FIGURE	DESCRIPTION
-MAS	FIG.1	MOTOR FOR SPRING CHARGING	-BGS2	FIG.26	SIGNALLING - SPRING CHAR./DISCHAR.
-BGS1	FIG.1	CONTACT FOR CHARGER MECHANISM			
-MBC	FIG.2	CLOSING RELEASE			
-RLE1	FIG.4	ELECTROMECHANICAL LOCK CB CLOSING			
			-BGB1	FIG.34	AUXILIARY SWITCH CB SHAFT
			-BGB1	FIG.34	AUXILIARY SWITCH CB SHAFT
-MBO1	FIG.7	FIRST OPENING RELEASE	-BGB1	FIG.34	AUXILIARY SWITCH CB SHAFT
-RLE2	FIG.8	ELECTROMECHANICAL LOCK CB TRUCK	-BGT1	FIG.52	CB TRUCK IN SERVICE (CONNECTED) POSITION
			-BGT2	FIG.52	CB TRUCK IN TEST (ISOLATED) POSITION
-XE		GROUND TERMINALS	-XDB		58-POLE PLUG FOR AUXILIARY CIRCUITS

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K01
name	name	name			Customer document number				
Revision / Date	Name	Remark	Document number	1VL84000001	Customer			Place of installation +	LV
			Date	date		Page description	Incoming Feeder VD4 INTERNAL WIRING	DCC &	&EFS
			ABB	ABB s.r.o. EPDS Brno				Page	M01
								Pages	28

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SAMPLE

+LVD-BCF1

REF615: FEEDER PROTECTION RELAY

REF615

Terminal	Signal	Cable	Function
-X120: 7	IL1	S01/7B	IL1 N
-X120: 8	IL1	S01/7B	
-X120: 9	IL2	S01/7C	IL2 N
-X120: 10	IL2	S01/7C	
-X120: 11	IL3	S01/7D	IL3 N
-X120: 12	IL3	S01/7D	
-X120: 13	I0		
-X120: 14	I0		
-X130: 9	NOT IN USE		
-X130: 10	NOT IN USE		
-X130: 11	U1	S17/2E	U1 N
-X130: 12	U1	S17/3E	
-X130: 13	U2	S17/3E	U2 N
-X130: 14	U2	S17/3E	
-X130: 15	U3	S17/3E	U3 N
-X130: 16	U3	S17/3E	
-X130: 17	U0	S17/5E	U0 N
-X130: 18	U0	S17/5E	
-X100: 1	PS	N03/3C	DC POWER SUPPLY (+/L)
-X100: 2	PS	N03/3D	DC POWER SUPPLY (-/N)
-X100: GND	GND	N03/3C	GND STUD
-X100: 3	IRF	X02/6D	IRF
-X100: 4	IRF	X02/6C	HEALTHY
-X100: 5	IRF	X02/6C	
-X100: 6	PO1	M06/4B	RELEASE FOR CB CLOSING (RLE1)
-X100: 7	PO1	M06/4B	
-X100: 8	PO2	M06/2B	COMMAND CB CLOSE (MBC)
-X100: 9	PO2	M06/2B	
-X100: 10	SO1	M06/5B	RELEASE FOR CB TRUCK (RLE2)
-X100: 11	SO1	M06/5B	
-X100: 12	SO1	M06/6B	
-X100: 13	SO2	M06/7B	RELEASE FOR ES (RLE3)
-X100: 14	SO2	M06/7B	
-X100: 16	PO3	M07/2B	COMMAND CB OPEN/TRIP (MBO1)
-X100: 17	PO3	M07/2B	
-X100: 15	TCS1	M07/2B	
-X100: 19	TCS1	M07/2C	
-X100: 18	PO4	M07/2C	
-X100: 20	PO4	M07/2C	
-X100: 22	TCS2		
-X100: 21	TCS2		
-X100: 23			
-X100: 24			
-X120: 1	BI1 COM	N06/2E	ITH LIMITOR ACTIVATED
-X120: 2	BI1 COM	N06/2E	
-X120: 3	BI2	N06/3E	SUPPLY MCBs OK
-X120: 4	BI3	N06/3E	METERING CIRCUIT MCBs OK
-X120: 5	BI4	N06/4E	SPARE
-X120: 6	BI4	N06/4E	
-X130: 1	BI1	N06/5E	SPARE
-X130: 2	BI1	N06/5E	
-X130: 3	BI2	N06/5E	SPARE
-X130: 4	BI2	N06/5E	
-X130: 5	BI3	N06/6E	SPARE
-X130: 6	BI3	N06/6E	
-X130: 7	BI4	N06/6E	SPARE
-X130: 8	BI4	N06/6E	
-X110: 1	BI1	N04/2E	STATUS CB OPEN
-X110: 2	BI1	N04/2E	
-X110: 3	BI2	N04/2E	STATUS CB CLOSE
-X110: 4	BI2	N04/2E	
-X110: 5	BI3 COM	N04/3E	CB TRUCK IN TEST
-X110: 6	BI3 COM	N04/3E	
-X110: 7	BI4	N04/3E	CB TRUCK IN SERVICE
-X110: 8	BI5 COM	N04/4E	STATUS ES CLOSE
-X110: 9	BI5 COM	N04/4E	
-X110: 10	BI6	N04/4E	ES OPEN AND CRANK NOT INSERTED
-X110: 11	BI7 COM	N04/4E	CB SPRING DISCHARGED
-X110: 12	BI7 COM	N04/4E	
-X110: 13	BI8	N04/5E	CB PLUG NOT INSERTED
-X110: 14	SO1		
-X110: 16	SO1		
-X110: 15	SO2		
-X110: 17	SO2		
-X110: 19	SO3		
-X110: 18	SO3		
-X110: 21	SO4		
-X110: 22	SO4		
-X110: 20			
-X110: 23			
-X110: 24			

REF615

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K01
name	name	name			Customer document number			Place of installation +	LV
Revision / Date	Name	Remark	Document number	1VL84000001	Customer	Page description	Incoming Feeder REF615 - ANALOG I/O BOARD	DCC &	&EFS
			Date	date				Page	N01
			ABB s.r.o. EPDS Brno					Pages	28

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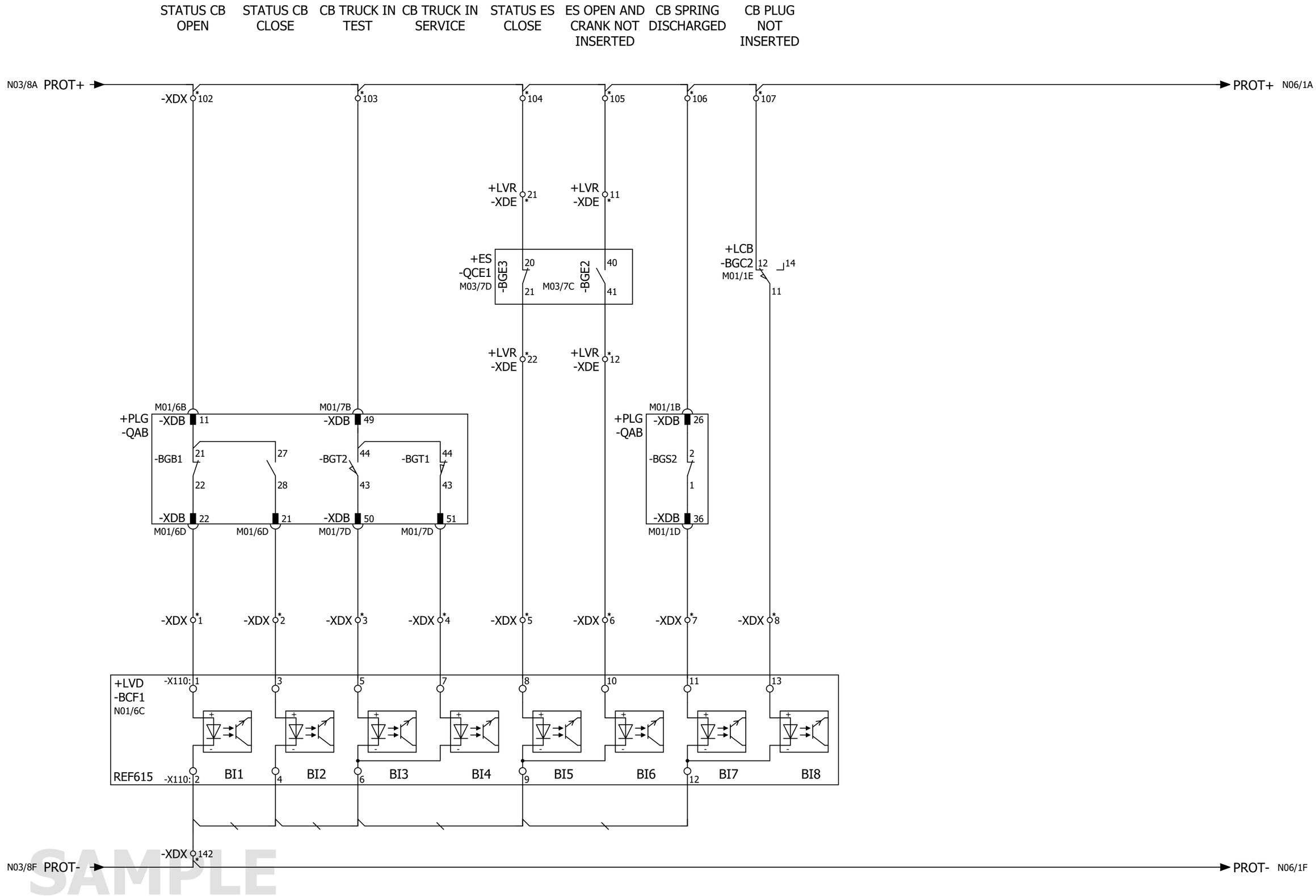
SAMPLE



SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K01
name	name	name			Customer document number				
Revision / Date	Name	Remark	Document number	1VL84000001	Customer	Page description	Incoming Feeder REF615 - BINARY INPUTS	Place of installation +	LV
			Date	date				DCC &	&EFS
			ABB s.r.o. EPDS Brno					Page	N04
								Pages	28

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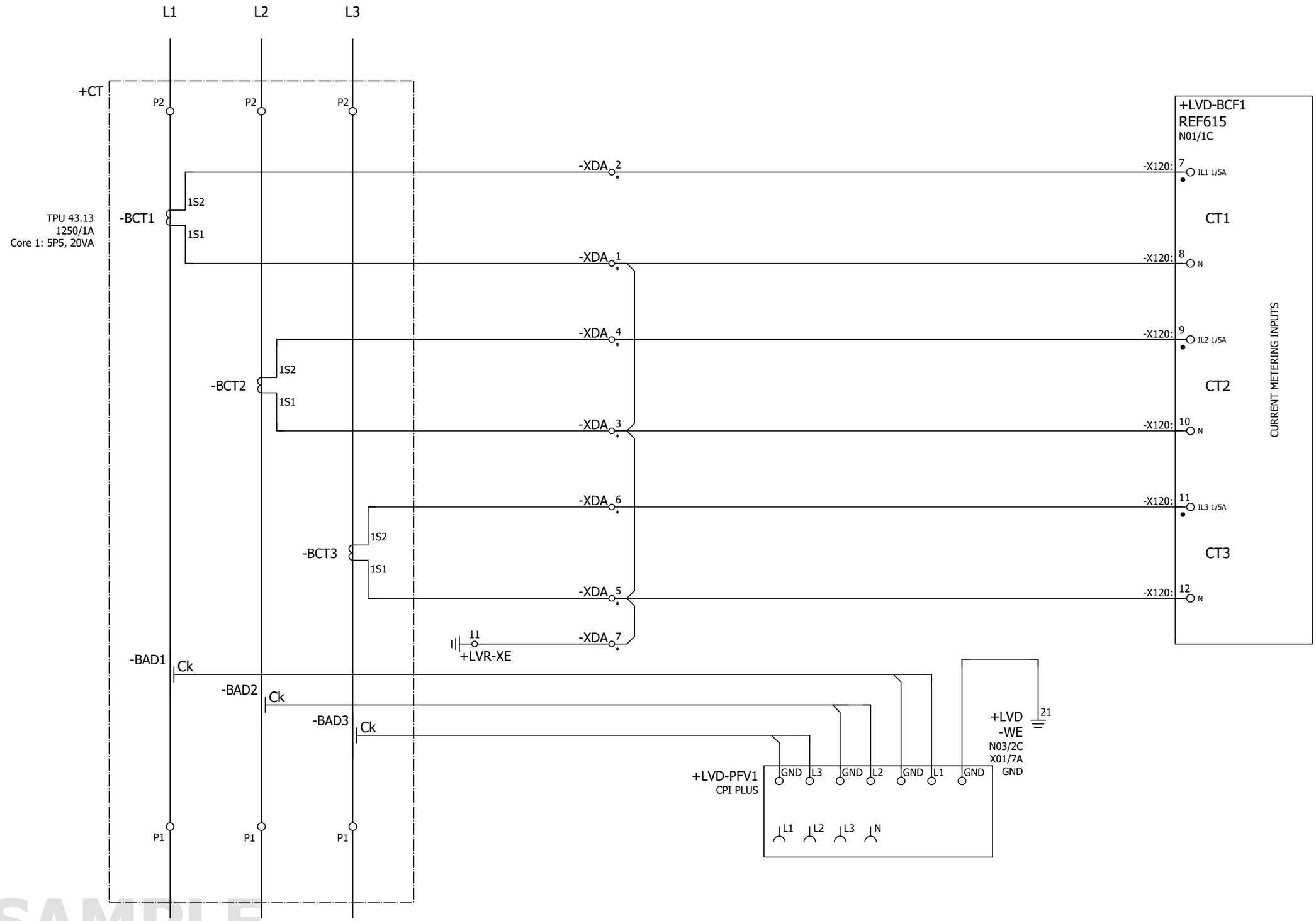
SAMPLE



SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K01
name	name	name			Customer document number				
Revision / Date	Name	Remark	Document number	1VL84000001	Customer	Page description	Incoming Feeder CURRENT TRANSFORMER CORE 1	Place of installation +	LV
			Date	date				DCC &	&EFS
			ABB s.r.o. EPDS Brno					Page	S01
								Pages	28

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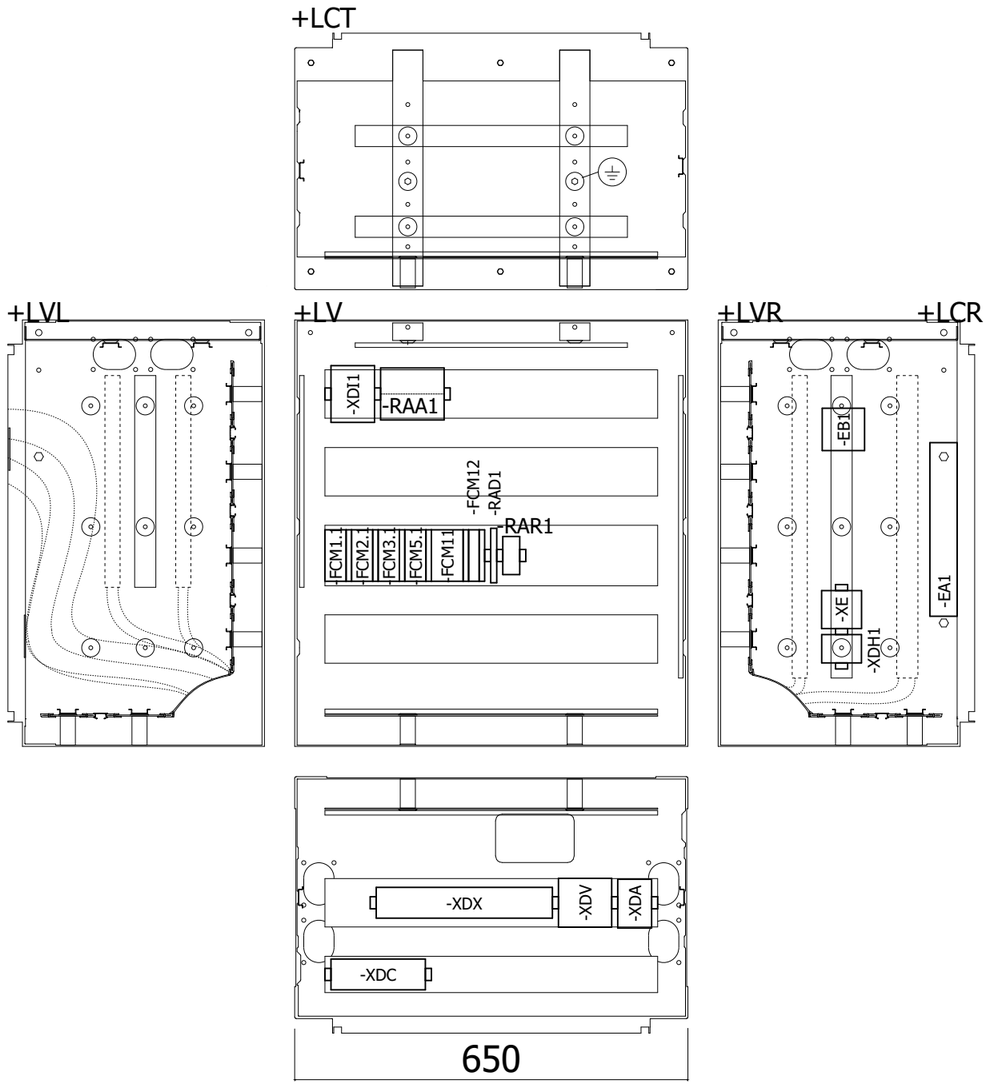
SAMPLE



ENCLOSURE LEGEND

=K01+LCR	
DEVICE TAG	TEXT
-EA1	LIGHT

=K01+LV	
DEVICE TAG	TEXT
-FCM1.1	PROTECTION CIRCUIT
-FCM2.1	CONTROL CIRCUIT
-FCM3.1	SPRING CHARGER CIRCUIT
-FCM5.1	HEATING & LIGHTING CIRCUIT
-FCM11	VT STAR WINDING 1
-FCM12	VT OEPN DELTA WINDING
-RAA1	ANTIFERRORESONANCE UNIT
-RAD1	DIODES
-RAR1	TCS MODULE

=K01+LVR	
DEVICE TAG	TEXT
-EB1	HEATER

Description of equipment (will be shown on the label)

Designation of equipment (will be shown on the label)

TERMINALS

=K01		
DT	TYPE	QTY
-XDA	TERMINAL ZS10-ST	7
-XDC	TERMINAL ZK2.5	30
-XDI1	TERMINAL ZDK4-22	12
-XDV	TERMINAL ZS10-ST	7
-XDX	TERMINAL ZK2.5	56
-XDE	TERMINAL ZK2.5-D2	10
-XDH	TERMINAL ZK2.5-D2	10
-XDH1	TERMINAL ZK2.5-S	9
-XE	TERMINAL ZK4-PE	11

SAMPLE

Drawn by	Checked by	Released by
name	name	name

Revision	Date	Name	Remark

Project number: 1098888888
Document number: 1VL84000001
Date: date

ABB s.r.o.
EPDS Brno

Customer project number
Customer document number
Customer

Project name: 12 kV Switchgear Sample
Page description: Incoming Feeder
PANEL LAYOUT

Installation =: K01
Place of installation +: LV
DCC &: &ELU
Page: V01
Pages: 28

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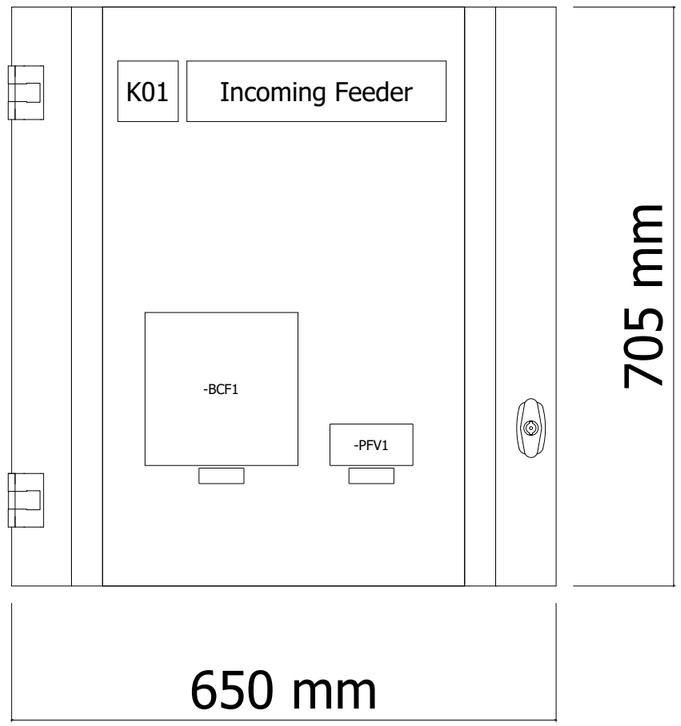
SAMPLE

ENCLOSURE LEGEND

=K01+LVD	
DEVICE TAG	TEXT
-BCF1	PROTECTION RELAY
-PFV1	MV VOLTAGE INDICATION

Description of equipment (will be shown on the label)

Designation of equipment (will be shown on the label)



SAMPLE

Drawn by	Checked by	Released by
name	name	name

Revision	Date	Name	Remark

Project number	1098888888
Document number	1VL84000001
Date	date

ABB s.r.o.
EPDS Brno

Customer project number
Customer document number
Customer

Project name: 12 kV Switchgear Sample

Page description: Incoming Feeder
LV COMP. DOOR LAYOUT

Installation = K01

Place of installation + LV

DCC &	Page	Pages
&ELU	W01	28

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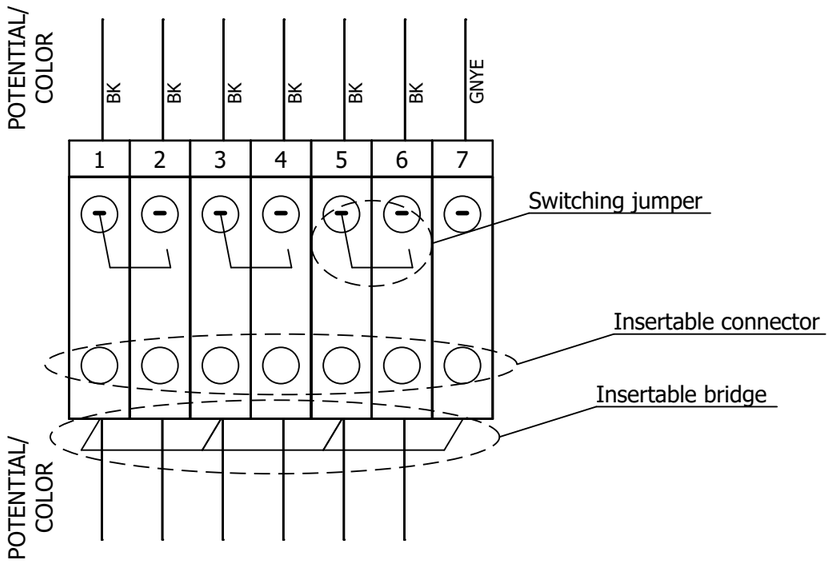
SAMPLE

TERMINALS
+LV-XDA

DEVICE DESIGNATION	DEVICE PIN	PAGE
+CT-BCT1	1S1	S01/4B
+CT-BCT1	1S2	S01/4B
+CT-BCT2	1S1	S01/4C
+CT-BCT2	1S2	S01/4C
+CT-BCT3	1S1	S01/4D
+CT-BCT3	1S2	S01/4D
+LVR-XE	11:1	S01/4D



DEVICE DESIGNATION	DEVICE PIN
+LVD-BCF1-X120	8
+LVD-BCF1-X120	7
+LVD-BCF1-X120	10
+LVD-BCF1-X120	9
+LVD-BCF1-X120	12
+LVD-BCF1-X120	11

TOTAL TERMINALS COUNT: 7 PCS
TERMINAL TYPE: TERMINAL ZS10-ST

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K01
name	name	name			Customer document number				
Revision / Date	Name	Remark	Document number	1VL84000001	Customer	Page description	Incoming Feeder	Place of installation +	LV
			Date	date			TERMINALS =K01+LV-XDA	DCC &	Page X10 / Pages 28
			ABB	ABB s.r.o. EPDS Brno					

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SAMPLE

TERMINALS +LV-XDX

	DEVICE DESIGNATION	DEVICE PIN	PAGE
210			M07/8A
211	+PLG-QAB-XDB	4	M06/2C
212	+PLG-QAB-XDB	10	M06/4C
213	+PLG-QAB-XDB	45	M06/5C
214	+LVR-XDE	31	M06/7C
215	-RAD1	1	M06/8C
220	+PLG-QAB-XDB	31	M07/2C
221	-RAR1	B	M07/4C
222	+PLG-QAB-XDB	41	M07/3D
223	+PLG-QAB-XDB	28	M07/3D
224	+PLG-QAB-XDB	38	M07/4E
291	-FCM2.1	2	M06/2E
292			M06/4E
293			M06/5E
294			M06/7E
295			M06/8E
296			M07/2E
297			M07/5E
298			M07/7E

POTENTIAL/COLOR

Insertable jumper

	DEVICE DESIGNATION	DEVICE PIN
210		
211	+LVD-BCF1-X100	9
212	+LVD-BCF1-X100	7
213	+LVD-BCF1-X100	10
214	+LVD-BCF1-X100	13
215		
220	+LVD-BCF1-X100	18
221		
222		
223		
224	-RAR1	G
291	+PLG-QAB-XDB	14
292	+PLG-QAB-XDB	20
293	+PLG-QAB-XDB	46
294	+LVR-XDE	32
295	-RAD1	2
296	+PLG-QAB-XDB	30
297		
298		

POTENTIAL/COLOR

TOTAL TERMINALS COUNT: 56 PCS
TERMINAL TYPE: TERMINAL ZK2.5

SAMPLE

Drawn by	Checked by	Released by	Project number	1098888888	Customer project number	Project name	12 kV Switchgear Sample	Installation =	K01
name	name	name			Customer document number				
Revision / Date / Name / Remark			Document number	1VL84000001	Customer	Page description	Incoming Feeder	Place of installation +	LV
			Date	date			TERMINALS =K01+LV-XDX	DCC &	Page X15 / Pages 28
			ABB s.r.o. EPDS Brno						

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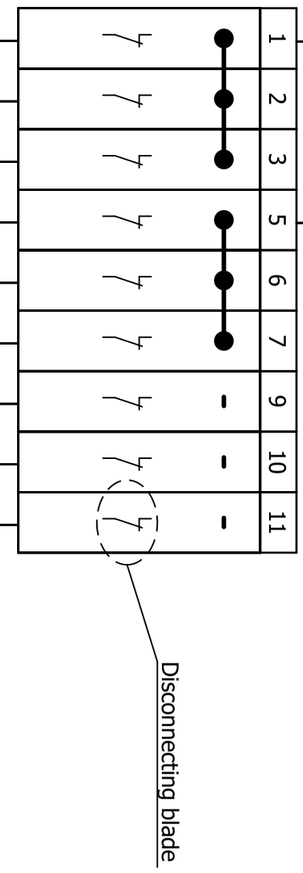
TERMINALS
+LVR-XDH1

DEVICE PIN	DEVICE DESIGNATION	POTENTIAL/ COLOR
BK	+CAB-BGA3	BK_ITH
BK	+CB-BGA1	BK_ITH
BK	+BB-BGA2	BK_ITH
BN	+CAB-BGA3	BN_ITH
BN	+CB-BGA1	BN_ITH
BN	+BB-BGA2	BN_ITH
GY	+CAB-BGA3	GY_ITH
GY	+CB-BGA1	GY_ITH
GY	+BB-BGA2	GY_ITH



POTENTIAL/ COLOR	DEVICE DESIGNATION	DEVICE PIN	PAGE
	+LV-XDX	151	+LVN06/2C
			+LVN06/1C
			+LVN06/1C
	+LV-XDX	152	+LVN06/2D
			+LVN06/2D
			+LVN06/1D
			+LVN06/2D
			+LVN06/1D
			+LVN06/1D

TOTAL TERMINALS COUNT: 9 PCS
TERMINAL TYPE: TERMINAL ZK2.5-S

SAMPLE

Drawn by name	Checked by name	Released by name	Project number 1098888888
Revision	Date	Name	Remark
			Document number 1VL84000001
			Date date
			ABB s.r.o. EPDS Brno

Customer project number
Customer document number
Customer

Project name: 12 kV Switchgear Sample

Page description: Incoming Feeder
TERMINALS =K01+LVR-XDH1

Installation = K01
Place of installation + LV
DCC &
Page X19
Pages 28